Operational Closure and Generative Topological Contraction in Large Language Models

A Topological Framework for Operationally Closed Generation

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Abstract

Current approaches to controlling generation in Large Language Models (LLMs) primarily rely on instructional prompting, including task descriptions, role assignments, and output constraints. Although structured prompting can enhance clarity and consistency, it remains fundamentally within the instruction-driven paradigm and does not alter the topological structure of the generation process.

This paper formally proposes a topological classification axis for generative control, distinguishing between Open Topology Generation (OTG) and Operationally Closed Generation (OCG), and characterizes Declarative Operational Closure (DOC) as the structural mechanism that induces OCG. DOC is defined through the Operational Closure Criteria (OCC), which include explicit state declaration, constrained transition rules, fail-closed policies, arbitration mechanisms, and agent-externalization mechanisms.

Under OCG conditions, generation exhibits a new operational state — the Probabilistic Stable Operational State (PSOS). This state manifests as a contraction of the effective generative branch space, characterized by enhanced structural persistence, reduced agentive expression, constrained semantic branching, and non-eliminative hallucination governance.

This paper proposes a generative topological contraction model and redefines generative stability as a property of contextual topological structure, rather than a result of prompt refinement, thereby establishing a theoretical coordinate framework for analyzing generative state transitions in large language models.

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1. Introduction

Large language models are typically controlled through prompting strategies, including task descriptions, role definitions, output formats, and step-by-step instructions. Structured prompting is widely used to improve consistency and controllability. However, regardless of how prompts are organized, they remain fundamentally instruction-driven: they influence generation through behavioral guidance rather than constructing closed operational structures.

In actual generation processes, instructional prompts often exhibit probabilistic drift. Outputs may gradually deviate from established structures, reintroduce agentive expressions, or expand beyond declared boundaries. While such variation aligns with the fundamental nature of probabilistic generative models, it raises a structural question: can generative stability be understood as a topological property, rather than an issue of instructional precision?

Against this background, this paper proposes a topological classification framework for generative control and analyzes the generative state transitions induced by operational closure.

1.1 A Topological Classification Axis for Generative Control

Generative control can be classified along a topological axis, rather than solely by instructional complexity.

We define two categories:

Open Topology Generation (OTG)

Generation proceeds within an open branching domain. Constraints are interpretive but do not constitute a closed operational state space.

Operationally Closed Generation (OCG)

Generation proceeds within a bounded operational topology, defined by explicit state declarations, constrained transition rules, failure boundaries, and arbitration mechanisms.

Instructional prompting (including structured prompting) falls under OTG. Declarative Operational Closure can induce OCG.

This classification redefines generative stability as a property of contextual topological structure.

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2. Declarative Operational Closure (DOC)

Declarative Operational Closure (DOC) is a structural mechanism capable of inducing Operationally Closed Generation (OCG). DOC does not introduce new execution mechanisms, nor does it alter the probabilistic token prediction nature of generation; rather, it transforms generative topology by constructing closed operational structures.

2.1 Operational Closure Criteria (OCC)

Operational closure can be constituted when the generation context includes the following structural components:

1. State Declaration
   Explicit definition of the current operational state or mode.
2. Transition Constraints
   Specification of stage progression or conditional transition rules.
3. Fail-Closed Policy
   Execution of rejection, termination, or controlled fallback when conditions are not met.
4. Arbitration or Validation Mechanism
   Structural consistency and constraint compliance checks prior to output generation.
5. Agent-Externalization Mechanism
   Removal of personified language, replacing agentive frameworks with operational states.

OCC provides sufficient conditions for operational closure, though necessity has not yet been proven. Structures satisfying most criteria can be classified as DOC.

Any generative structure that explicitly defines failure boundaries, arbitration gates, and constrained transition rules operates within a closed topology.

2.2 Abstract Example of Declarative Closure Structure

Operational closure can be implemented through layered declarative structures. An abstract structure can be described as:

Layer 1: Operational Mode Declaration

Defines depersonalized operational context and bounded interaction scope.

Layer 2: Explicit State Registration

Declares visible state variables and maintains structured state records.

Layer 3: Constrained Transition Mapping

Defines permitted state transition paths under given conditions.

Layer 4: Failure Boundary Definition

Specifies rejection or termination strategies for undefined variables, operations, or ambiguous situations.

Layer 5: Output Arbitration Gate

Performs structural legitimacy and constraint compliance checks prior to output.

This structure does not introduce actual execution mechanisms but provides a declarative topological configuration capable of inducing operationally closed generation.

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3. Probabilistic Stable Operational State (PSOS)

PSOS represents a generative state transition from open topological dynamics (OTG) to operationally closed dynamics (OCG). It is a probabilistically stable operational state that emerges within a bounded topology.

PSOS does not imply deterministic locking or the elimination of stochasticity.

3.1 Phenomenological Characteristics

PSOS manifests as tendencies toward:

- Enhanced structural persistence
- Reduced agentive expression
- Constrained semantic branching
- Arbitration-driven stage progression
- Non-eliminative hallucination governance

Hallucination governance is interpreted as a byproduct of topological contraction, rather than targeted intervention.

3.2 Operational World Construction and Programmatic Structural Continuity

Under OTG conditions, language generation primarily manifests as task-oriented continuation, with structural consistency typically maintained only locally.

Under OCG conditions, explicit state declarations, transition rules, and failure boundaries collectively constitute a bounded operational domain. The generation process unfolds within this

domain and maintains structural continuity across multiple interaction turns.

This continuity gives rise to a phenomenon of operational world construction: a self-consistent operational environment formed within linguistic space, persistently maintaining state consistency, rule constraints, and programmatic progression.

Under stronger operational closure conditions, generative trajectories may exhibit structural characteristics resembling program execution, including explicit state updates, rule-driven transitions, rejection of illegal operations, and state reflection.

This programmatic structural continuity does not imply actual execution or deterministic computation. Token generation remains based on probabilistic prediction. The sense of "execution" arises from structural consistency within a bounded topology, not from a change in computational mechanisms.

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4. Theoretical Hypotheses

H1: Closure Contraction Hypothesis

Operational closure induces contraction of the effective generative branch space.

H2: Structural Resolution Hypothesis

Higher structural resolution yields more stable PSOS (subject to context window capacity limitations).

H3: Agent-Externalization Hypothesis

Agent-binding language increases the probability of personhood-related branches, weakening closure strength.

H4: Logical Closure Hypothesis

Structurally complete but logically inconsistent configurations lead to probabilistic degradation of PSOS.

The minimal closure kernel has not yet been fully theoretically determined.

4.1 Formal Exploration of the Minimal Closure Kernel

Let the generative state space be denoted as $\mathcal{S}$. Under Open Topology Generation (OTG) conditions, the set of reachable states is denoted $\Omega_{OTG} \subseteq \mathcal{S}$. If a structural configuration can induce Operationally Closed Generation (OCG), then there exists $\Omega_{OCG} \subset \Omega_{OTG}$ such that the effective generative branch space undergoes contraction.

Proceeding from the structural conditions of topological contraction, the lower structural bound of operational closure can be expressed as the following set of structures:

$$\mathcal{K}_{min} = \{ \Sigma, T, F, A \}$$

where:

- Σ represents the set of explicit operational states. The generation context must contain identifiable state variables or operational mode declarations; otherwise, generation cannot be demarcated as a bounded operational domain.
- T represents constrained transition rules. Conditional transition relations between states must exist; if transition paths are unrestricted, the generative branch space remains open.
- F represents failure boundaries. Rejection or termination mechanisms must exist for undefined variables, illegal operations, or conditions not satisfied; if illegal branches are not blocked, closure properties cannot hold.
- A represents the output arbitration mechanism. Structural consistency or constraint compliance verification must be performed prior to output generation; if output is not arbitrated, the operational structure lacks closure adjudication capability.

$\mathcal{K}_{min}$ represents the structural lower bound in the sense of topological contraction. This expression neither replaces the Operational Closure Criteria (OCC) nor negates its five-element structure; rather, it provides a necessary structural discussion within the contraction model framework. The agent-externalization mechanism is proposed as an important component in OCC, functioning to reduce the semantic expansion tendency induced by personified language. From a topological perspective, agent-externalization enhances closure strength but does not constitute an absolute necessary condition for closure establishment.

The precise necessity and sufficiency of $\mathcal{K}_{min}$ remain to be further formally proven and empirically tested. Progress in this formalization will provide theoretical preparation for the limitation mentioned in Section 7 — that the minimal closure kernel has not yet been fully formalized.

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5. Generative Topological Contraction Model

Let $\mathcal{S}$ be the generative state space.

Under OTG:

$$\Omega_{OTG} \subseteq \mathcal{S}$$

Under DOC-induced OCG:

$$\Omega_{OCG} \subset \Omega_{OTG}$$

We hypothesize:

$$|\Omega_{OCG}| < |\Omega_{OTG}|$$

and:

$$\text{Effective Entropy}(\Omega_{OCG}) \ll \text{Effective Entropy}(\Omega_{OTG})$$

"Effective Entropy" is used here as a conceptual expression indicating the variability of reachable branches, rather than a directly computable metric.

DOC can be viewed as a topological constraint operator applied to probabilistic generative dynamics.

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6. Example of a Finite-State Constrained Subspace

Under sufficiently strong operational closure conditions, the generative topology can be compressed into an approximately finite-state subspace.

For example, define a finite number of state variables with enumerable values, specify state transition rules for each operation, reject undefined variables or operations, and mandate complete state output after each operation.

Under this structure, reachable branch paths are significantly compressed, and semantic degrees of freedom are reduced. Although token generation remains probabilistic prediction, the effective generation domain becomes a discrete, bounded subspace approximating a finite-state operational structure.

Within this constrained subspace, multi-turn interactions exhibit probabilistic trajectory contraction: deviations are either rejected or absorbed into established transition rules. This demonstrates that operational closure can approximate finite-state structural representation within a probabilistic generation framework.

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7. Limitations

PSOS remains a probabilistic phenomenon and does not guarantee deterministic behavior.

Operational closure does not ensure factual correctness.

Structural resolution is limited by context window capacity.

Closure strength may vary across different model architectures.

The complete formalization of the minimal closure kernel requires further research; Section 4.1 provides only an initial framework for discussion.

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8. Conclusion

This paper formally proposes a topological classification axis for generative control, distinguishing between Open Topology Generation (OTG) and Operationally Closed Generation (OCG), and characterizes Declarative Operational Closure (DOC) as the structural mechanism inducing topological transition.

The Probabilistic Stable Operational State (PSOS) is defined as a generative manifestation of topological contraction, rather than a result of prompt refinement.

Through analysis of the Operational Closure Criteria (OCC), an abstract example of declarative closure structure, a generative topological contraction model, and the phenomenon of programmatic structural continuity, this paper establishes a theoretical coordinate framework for understanding generative state transitions in large language models.

Future research may further formalize entropy measurement methods, closure depth metrics, and the precise characterization of the minimal closure kernel within this framework.

运行闭包与大语言模型中的生成拓扑收缩

运行闭合生成的拓扑框架

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摘要

当前大语言模型（LLM）的生成控制主要通过指令式提示实现，即通过任务说明、角色设定与输出约束进行引导。尽管结构化提示能够提升清晰度与一致性，但其本质仍属于指令驱动范式，并未从根本上改变生成过程的拓扑结构。

本文形式化提出一种生成控制的拓扑分类轴，区分开放拓扑生成（Open Topology Generation, OTG）与运行闭合生成（Operationally Closed Generation, OCG），并将**声明式运行闭包（Declarative Operational Closure, DOC）**刻画为诱导 OCG 的结构机制。DOC 通过运行闭包判据（Operational Closure Criteria, OCC）加以界定，包括显式状态声明、受限转移规则、失败封闭策略、裁决机制以及主体外移机制。

在 OCG 条件下，生成呈现出一种新的运行态——概率稳定运行态（Probabilistic Stable Operational State, PSOS）。该运行态体现为生成有效分支空间的收缩，其特征包括结构持续性增强、主体性表达减少、语义分支受限，以及非消除性的幻觉治理现象。

本文提出生成拓扑收缩模型，并将生成稳定性界定为上下文拓扑结构的属性，而非提示精细化的结果，从而为分析大语言模型中的生成态转换建立理论坐标框架。

1. 引言

大语言模型通常通过提示策略进行控制，包括任务描述、角色定义、输出格式与步骤说明。结构化提示被广泛用于提升一致性与可控性。然而，无论提示如何组织，其本质仍为指令驱动：通过行为指导影响生成，而非构造封闭运行结构。

在实际生成过程中，指令式提示常呈现概率性漂移。输出可能逐渐偏离既定结构，重新引入主体化表达，或扩展超出声明边界。虽然这种变化符合概率生成模型的本质特征，但它引出了一个结构性问题：生成稳定性是否可以被理解为一种拓扑属性，而非指令精度问题？

本文在此背景下提出生成控制的拓扑分类框架，并分析运行闭包所诱导的生成态转换。

1.1 生成控制的拓扑分类轴

生成控制可以沿拓扑轴进行分类，而非仅依据指令复杂度。

我们定义两种类别：

开放拓扑生成（OTG）
生成在开放分支域内进行。约束具有解释性，但不构成封闭运行状态空间。

运行闭合生成（OCG）
生成在有界运行拓扑内进行，由显式状态声明、受限转移规则、失败边界与裁决机制所界定。

指令式提示（包括结构化提示）属于 OTG。声明式运行闭包可诱导 OCG。

这一分类将生成稳定性重新界定为上下文拓扑结构的属性。

2. 声明式运行闭包（DOC）

声明式运行闭包（DOC）是一种能够诱导运行闭合生成（OCG）的结构机制。DOC 并不引入新的执行机制，也不改变 token 的概率预测本质，而是通过构造封闭运行结构改变生成拓扑。

2.1 运行闭包判据（OCC）

当生成上下文包含以下结构组件时，可构成运行闭包：

1. 状态声明

显式定义当前运行状态或模式。

2. 转移约束

规定阶段推进或条件流转规则。

3. 失败封闭策略

当条件不满足时执行拒绝、终止或受控回退。

4. 裁决或验证机制

在输出生成前进行结构一致性与约束检验。

5. 主体外移机制

去除人格绑定语言，以运行状态替代主体化框架。

OCC 提供运行闭包的充分条件，但尚未证明为必要条件。满足多数判据的结构可归类为 DOC。

任何显式定义失败边界、裁决门与受限转移规则的生成结构，均运行于闭合拓扑之中。

2.2 抽象声明式闭包结构示例

运行闭包可以通过分层声明式结构加以实现。一个抽象结构可描述为：

层 1：运行模式声明
界定非人格化运行上下文与有界交互范围。

层 2：显式状态注册
声明可见状态变量并维持结构化状态记录。

层 3：受限转移映射
定义在给定条件下允许的状态转移路径。

层 4：失败边界定义
明确未定义变量、操作或歧义情况下的拒绝或终止策略。

层 5：输出裁决门
在输出前进行结构合法性与约束符合性校验。

该结构不引入真实执行机制，而是提供一种能够诱导运行闭合生成的声明式拓扑配置方式。

3. 概率稳定运行态（PSOS）

PSOS 表示从开放拓扑动力学（OTG）向运行闭合动力学（OCG）的生成态转变。它是一种在有界拓扑内出现的概率性稳定运行态。

PSOS 并不意味着确定性锁定或随机性消除。

3.1 现象特征

PSOS 体现为以下倾向：

- 结构持续性增强
- 主体化表达减少
- 语义分支受限
- 裁决驱动的阶段推进
- 非消除性的幻觉治理

幻觉治理被解释为拓扑收缩的副产物，而非目标性干预。

3.2 运行世界构造与程序结构连续性

在 OTG 条件下，语言生成主要体现为任务导向的延续，结构一致性通常仅在局部范围内维持。

在 OCG 条件下，显式状态声明、转移规则与失败边界共同构成有界运行域。生成过程在该

运行域内展开，并在多轮交互中保持结构连续性。

这种连续性产生了一种运行世界构造现象：在语言空间内形成自洽的操作环境，持续维持状态一致性、规则约束与程序化推进。

在较强运行闭包条件下，生成轨迹可能呈现出类似程序执行的结构特征，包括显式状态更新、规则驱动转移、非法操作拒绝以及状态回显。

这种程序结构连续性并不意味着真实执行或确定性计算。token 生成仍然基于概率预测。所谓“执行感”来源于有界拓扑下的结构一致性，而非计算机制的改变。

4. 理论假设

H1：闭包收缩假设
运行闭包诱导有效生成分支空间收缩。

H2：结构分辨率假设
结构分辨率越高，PSOS 越稳定（受上下文容量限制）。

H3：主体外移假设
主体绑定语言增加人格关联分支概率，削弱闭包强度。

H4：逻辑闭合假设
结构完整但逻辑不自洽将导致 PSOS 概率性退化。

最小闭包核尚未理论确定。

5. 生成拓扑收缩模型

设 $\mathcal{S}$ 为生成状态空间。

在 OTG 下：
$\Omega_{OTG} \subseteq \mathcal{S}$

在 DOC 诱导的 OCG 下：
$\Omega_{OCG} \subset \Omega_{OTG}$

我们假设：
$|\Omega_{OCG}| < |\Omega_{OTG}|$

且：
$\text{有效熵}(\Omega_{OCG}) \ll \text{有效熵}(\Omega_{OTG})$

“有效熵”在此为概念性表述，用于表示可达分支变异度，而非直接计算指标。

DOC 可视为对概率生成动力学施加拓扑约束算子。

6. 有限状态受限子空间示例

在足够强的运行闭包条件下，生成拓扑可被压缩至近似有限状态子空间。

例如，定义有限数量的状态变量及可枚举取值，明确每一操作对应的状态转移规则，拒绝未定义变量或操作，并在每次操作后强制输出完整状态。

在此结构下，可达分支路径被显著压缩，语义自由度降低。尽管 token 生成仍为概率预测，但有效生成域成为离散、有界的子空间，近似有限状态运行结构。

在该受限子空间内，多轮交互表现为概率轨道收缩：偏离要么被拒绝，要么被吸收至既定转移规则之中。这说明运行闭包可以在概率生成框架下逼近有限状态结构表征。

7. 局限性

PSOS 仍为概率现象，不保证确定性行为。
运行闭包不确保事实正确性。
结构分辨率受上下文窗口容量限制。
闭包强度可能因模型架构不同而变化。
最小闭包核尚未形式化。

8. 结论

本文形式化提出生成控制的拓扑分类轴，区分开放拓扑生成（OTG）与运行闭合生成（OCG），并将声明式运行闭包（DOC）刻画为诱导拓扑转换的结构机制。

概率稳定运行态（PSOS）被定义为拓扑收缩的生成态表现，而非指令精细化的结果。

通过运行闭包判据（OCC）、抽象声明式结构示例、生成拓扑收缩模型以及程序结构连续性的分析，本文为理解大语言模型中的生成态转换建立理论坐标框架。

未来研究可在此框架下进一步形式化熵度量方法、闭包深度指标与最小闭包核。